The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 September 2009.

INCH-POUND

MIL-PRF-19500/624D
18 June 2009
SUPERSEDING
MIL-PRF-19500/624C
2 June 2008

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, HIGH-POWER, TYPE 2N7370, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon high-power Darlington transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-254).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$ (2)
	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N7370	100	100	100	5.0	0.2	12	-65 to +200	1.5

(1) See figure 2 for temperature-power derating curves.
(2) See figure 3, transient thermal impedance graph.

1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

Limit	h_{FE1} (1) V_{CE} = 3.0 V dc I_C = 6.0 A dc	$V_{BE(SAT)1}$ (1) I_C = 12.0 A dc I_B = 120 mA dc	$V_{CE(SAT)1}$ (1) I_C = 12.0 A dc I_B = 120 mA dc	$\|h_{fe}\|$ V_{CE} = 3.0 V dc I_C = 5.0 A dc f = 1 MHz
		V dc	V dc	
Min	1,000			10
Max	18,000	4.0	3.0	250

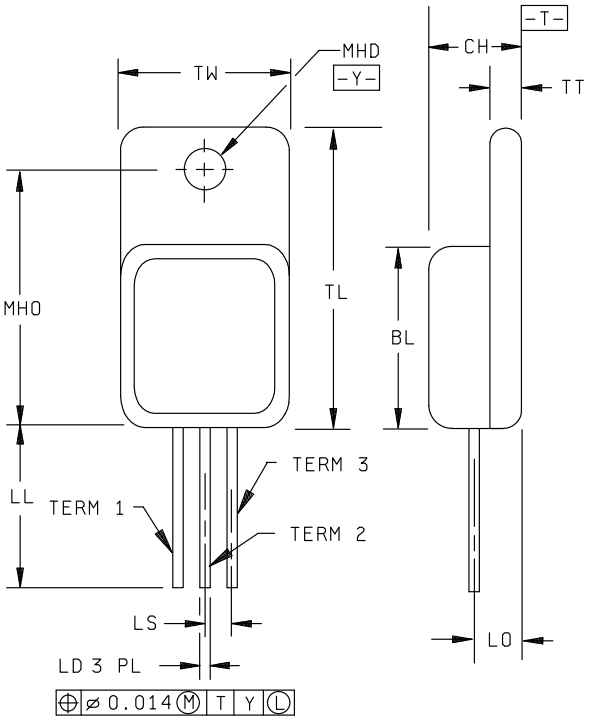
(1) Pulsed see 4.5.1.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

AMSC N/A

FSC 5961

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84
Term 1	Base			
Term 2	Collector			
Term 3	Emitter			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration (TO-254AA).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1. Methods used for electrical isolation of the terminal feedthroughs shall employ material that contain a minimum of 90 percent ceramic AL_2O_3 or equivalent. Examples of such construction techniques are metalized eyelets or ceramic walled packages.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2). When lead information is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with appendix E, table IV, screen 14, of MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (list applicable JAN levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750. (see 4.3.2)	Thermal impedance (transient), method 3131 of MIL-STD-750. (see 4.3.2)
9	I_{CEX1} and h_{FE2}	Not applicable
11	Subgroup 2 of table I herein; I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100 percent of initial value or 2 μA dc, whichever is greater. Δh_{FE2} = ± 40 percent of initial value.	I_{CEX1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100 percent of initial value or 2 μA dc, whichever is greater. Δh_{FE2} = ±40 percent of initial value.	Subgroup 2 of table I herein; I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100 percent of initial value or 2 μA dc, whichever is greater. Δh_{FE2} = ±40 percent of initial value.

* (1) This test shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = +175°C min, V_{CE} = 10-30 V dc, T_A = +30 ±5°C.

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4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, V_H, and t_{MD} (and V_C where appropriate). The $Z_{\theta JX}$ limit used in screen 3c shall comply with the thermal impedance graph on figure 3 (less than or equal to the curve value at the same t_H time) and/or shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E; table V of MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS) and VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with the applicable steps of table III herein.

* 4.4.2.1 Group B inspection (JANS), table E-VIA of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	$V_{CB} \geq 10$ V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	2037	Bond strength, test condition A

4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
* B3	1037	$V_{CB} \geq 10$ V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B6	1032	T_A = +200°C.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Conditions
C2	1056	Test condition B.
C2	2036	Test condition A (tension), weight = 4.5 kg, t = 10 seconds.
C5	3131	See 4.3.2.
* C6	1037	$V_{CB} \geq 10$ V dc, 6,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.

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* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 1						
	Visual and mechanical examination	2071					
	Subgroup 2						
*	Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JC}$			
	Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{CEO(sus)}$	100		V dc
	Collector-emitter cutoff current	3036	Bias condition D; V_{CE} = 50 V dc	I_{CEO}		1.0	mA dc
	Emitter-base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		2.0	mA dc
	Collector-emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc; V_{CE} = 100 V dc	I_{CEX1}		0.01	mA dc
	Base-emitter saturated voltage	3066	Test condition A; I_C = 12 A dc; I_B = 120 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		4.0	V dc
	Collector-emitter saturated voltage	3071	I_C = 12 A dc; I_B = 120 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)}$		3.0	V dc
	Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc; I_C = 6.0 A dc; pulsed (see 4.5.1)	h_{FE1}	1,000	18,000	
	Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc; I_C = 12 A dc; pulsed (see 4.5.1)	h_{FE2}	150		
	Subgroup 3						
	High-temperature operation:		T_A = +150°C				
	Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 100 V dc; V_{BE} = 1.5 V dc	I_{CEX2}		5.0	mA dc
	Low-temperature operation:		T_A = -55°C				
	Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc; I_C = 6.0 A dc; pulsed (see 4.5.1)	h_{FE3}	300		

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 4						
	Switching parameters						
	Turn-on		See figure 4	t_{on}		2.0	µs
	Turn-off		See figure 4	t_{off}		10	µs
	Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 3.0 V dc; I_C = 5.0 A dc; f = 1 MHz	$\|h_{fe}\|$	10	250	
	Subgroup 5						
	Safe operating area (continuous dc)	3051	T_C = +25°C; t ≥ 1 s; 1 cycle; (see figure 5)				
	Test 1		V_{CE} = 8.3 V dc; I_C = 12.0 A dc				
	Test 2		V_{CE} = 30 V dc; I_C = 3.3 A dc				
	Test 3		V_{CE} = 90 V dc; I_C = 150 mA dc				
	Safe operating area (clamped inductive)	3053	Load condition B (clamped inductive load); T_A = +25°C; $t_r + t_f \leq 1.0$ µs; duty cycle ≤ 2 percent; t_p = 1 ms; (vary to obtain I_C); R_S = 0.10 Ω; R_{BB1} = 80 Ω; V_{BB1} =16 V dc; R_{BB2} = 100 Ω; V_{BB2} = 1.5 V dc; I_C =12 A dc; V_{CC} = 20 V dc; R_L ≤2 Ω; L = 10 mH; (Stancor C-2688 or equivalent) clamp voltage = 100 +0, -5 V dc; Device fails if clamp voltage not reached.				
*	Electrical measurements		See table I, subgroup 2 herein.				
	Subgroups 6 and 7						
	Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

	Inspection	MIL-STD-750		Qualification conformance inspection
		Method	Conditions	
	Subgroup 1			45 devices c = 0
	Temperature cycling	1051	500 cycles	
	Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
*	Electrical measurements		See table I, subgroup 2.	
	Subgroup 2			45 devices c = 0
	High temperature reverse bias	1039	Condition A; 1,000 hours	
*	Electrical measurements		See table I, subgroup 2.	
	Subgroup 4			
	Thermal impedance curves		See MIL-PRF-19500	
*	Subgroup 5			
	Not applicable			
	Subgroup 8			45 devices c = 0
*	Reverse stability	1033	Condition B.	

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* TABLE III. Group B delta measurements. 1/

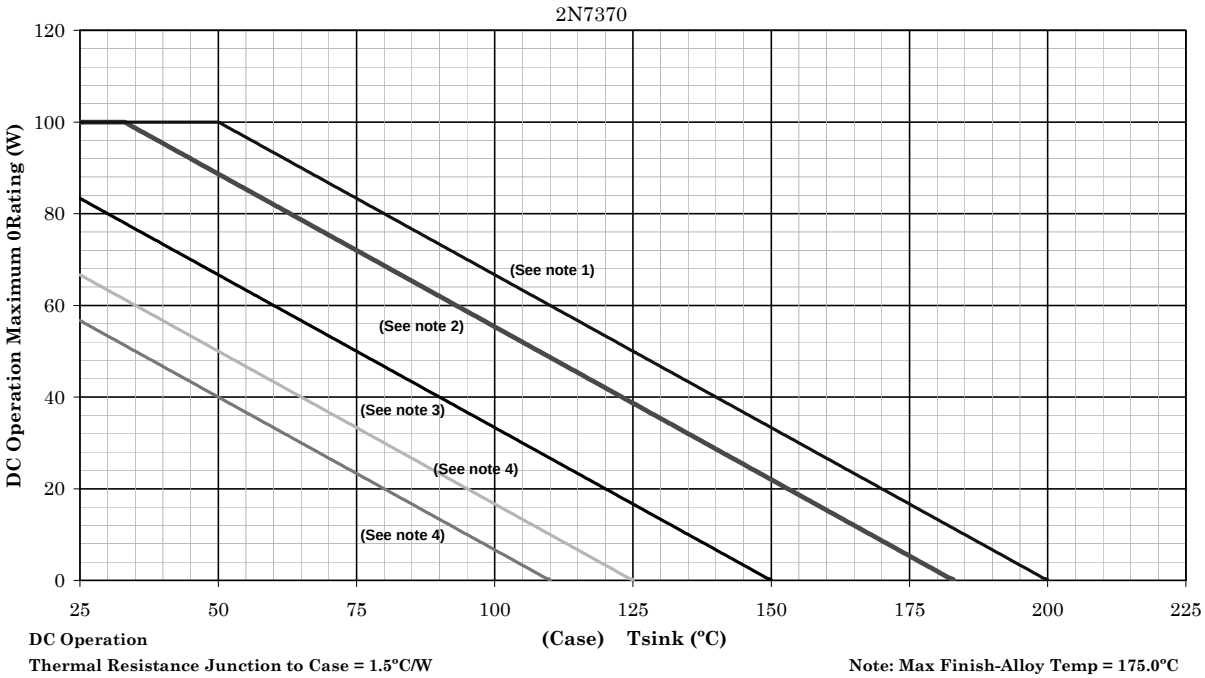
Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc; I_C = 6.0 A dc; pulsed (see 4.5.1)	Δh_{FE1} 2/	±40 percent change from initial value.		

1/ The delta measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
a. Subgroup 4, see table II herein, step 1
b. Subgroup 5, see table II herein, step 1.

2/ Devices which exceed the group A limits for this test shall not be shippable but are not considered failures for the test.

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Temperature-Power derating curve



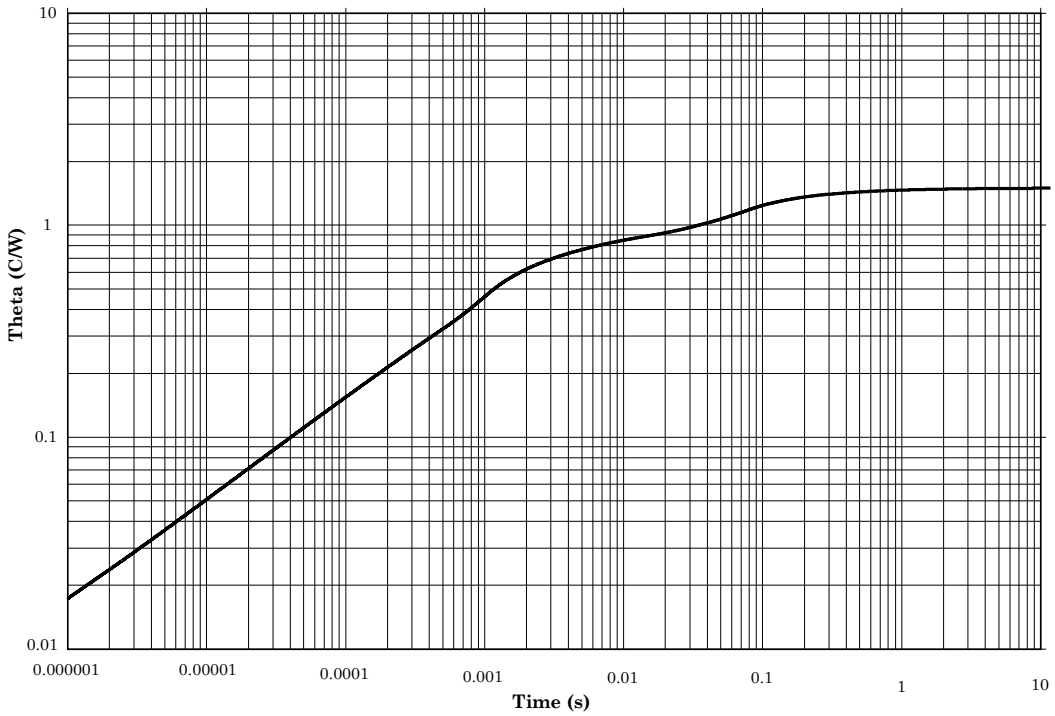
NOTES:

1. Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Temperature-power derating graph.

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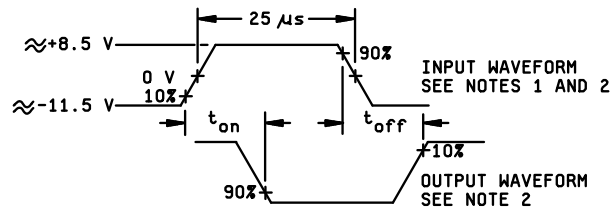
Maximum Thermal Impedance

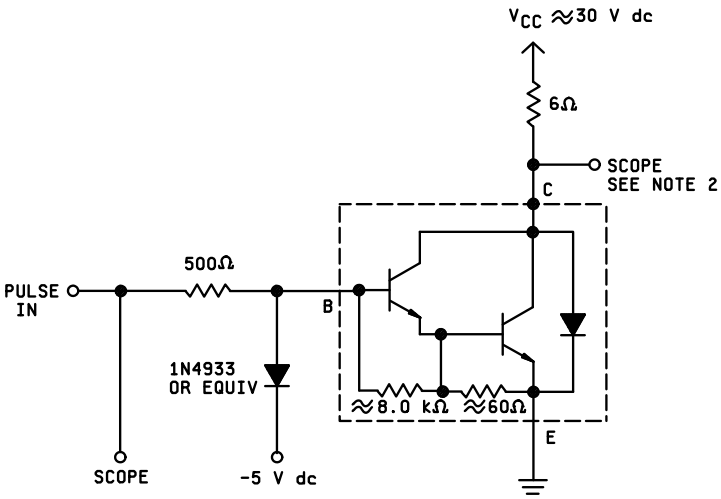


T_C = +25C. Thermal impedance = 1.5°C/W.

* FIGURE 3. Transient thermal impedance graph.

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NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20$ ns, $t_f \leq 20$ ns, $Z_{OUT} = 50\ \Omega$, PW = 25 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 20$ ns, $Z_{IN} \geq 20\ k\Omega$, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 4. Pulse response test circuit.

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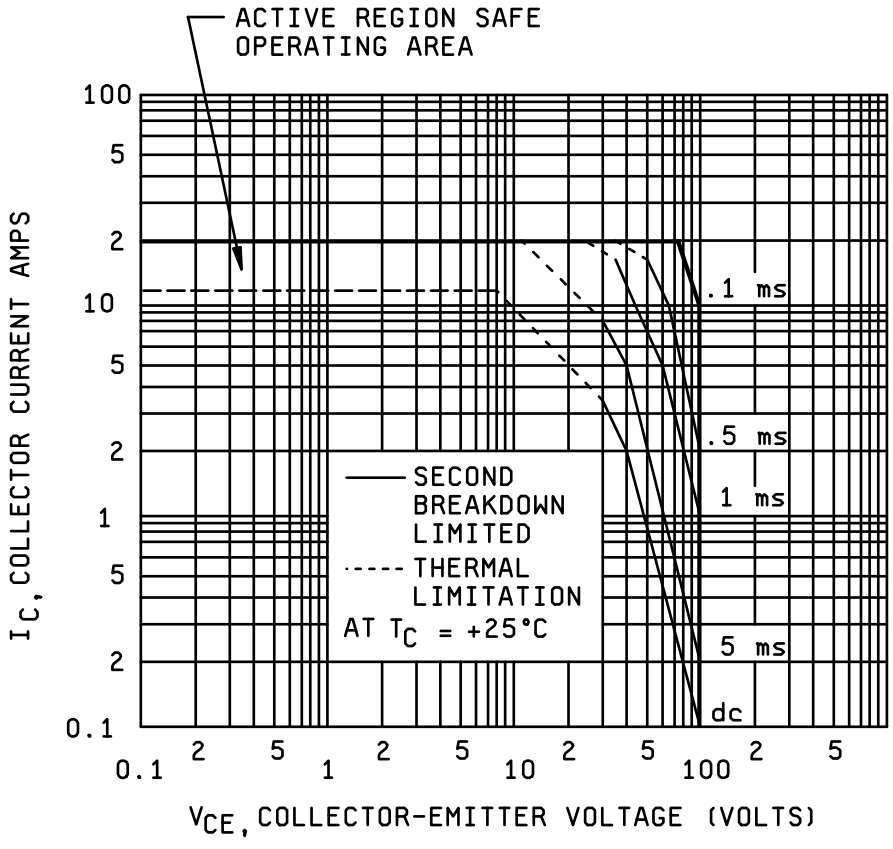


FIGURE 5. Safe operating area.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

6.4 Interchangeability information. MIL-PRF-19500/624 is a TO-254 package version of MIL-PRF-19500/502, which is a TO-3 package version. The military 2N7370 contains the same die as the military 2N6059. The MIL-PRF-19500/624 is preferred over the MIL-PRF-19500/502 whenever interchangeability is not a problem. For new design use 2N7370. The 2N6059 is inactive for new design.

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS
Air Force - 19, 99

Preparing activity:
DLA - CC

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